

CS338 - Introduction to the Theory of Computation

► Revision class

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➤ From Lecture 0:

At the end of this course you'll be able to...

- Analyse rigorously whether a computational problem is **solvable** or not by different computational models
- Describe the **classes of problems** that different models of computation can solve
- Provide examples of problems that **no** reasonable computational model can solve
- Show whether a computational problem can be solved **efficiently**
- Show when a problem is **at least as hard** as another
- Prove to which **computational complexity class** a problem belongs to (i.e., how hard the problem is).

➤ What you've learned (1)

- **Theoretical foundations** of computer science
 - Formal definitions of everyday concepts
 - What is a computer?
 - What is an algorithm?
 - Determinism and nondeterminism
 - Concepts that may not be practical can help us understand practical things
 - Proof techniques
 - Induction
 - Diagonalisation
 - By closure properties
 - By contradiction

➤ What you've learned (2)

- Computability theory **(Part I)**
 - **Understanding** "what is computable"?
 - What problems can be solved in principle and which cannot be solved independent from the amount of available resources
 - Decidable / undecidable - recognisable/unrecognisable problems
 - **Computational models of increasing complexity**
 - Finite state automata
 - Pushdown automata
 - 1-tape Turing machines
 - More complex Turing machines (multi-tape, enumerators)

➤ What you've learned (3)

- **How to classify different computational models according to their power**
 - $\text{DFA} \Leftrightarrow \text{NFA} \rightarrow$ Regular languages, regular expressions ($\cup, ^\circ, *$)
 - $\text{PDA} \Leftrightarrow \text{DFA with a stack} \rightarrow$ Context Free Languages, CFG ($S \rightarrow R, R \rightarrow a$)
 - $\text{DTM} \Leftrightarrow \text{NTM} \Leftrightarrow$ Any computer $\rightarrow$ decidable and recognisable languages
- **How to show that a computational model is not powerful enough**
 - Pumping Lemmas (Not regular, not CFL)
 - Reductions
 - General: Not decidable
 - Mapping: Not Turing recognisable

➤ What you've learned (4)

- Computational complexity (Part II)
 - How to classify problems according to their difficulty
 - P Class, NP Class, co-NP class, NP-Complete class, NP-hard
 - How to decide whether a problem is harder than another
 - *What makes problems computationally harder than others?*
 - Is it in P, NP, co-NP, NP-Complete, NP-Hard, PSPACE. EXPTIME?

➤ What you've learned (5)

- Some of the most important **open** problems in computer science
 - P vs NP problem,
 - where to place eg., Factorisation?
- Some of the **foundational concepts** in computer science
 - Church-Turing thesis
 - Cook-Levin Theorem (SAT is NP-complete)
 - Rice's Theorem
 - Halting problem
 - Etc.

➤ What you've learned (6)

- Some of the **most studied** combinatorial optimisation problems (and why):
 - SAT, SUBSET SUM
 - HAMPATH, CLIQUE, VERTEX COVER, GRAPH COLORING, TSP
- Applied (revised) some **algorithmic design techniques**:
 - Graph searching (BFS)
 - Greedy algorithms (2-Graph colouring)
 - Dynamic Programming (CFL are decidable)
- Motivation for other **advanced courses**:
 - CS208 (Algorithms Design and Analysis),
 - CSE5003/CS5018 (Advanced Algorithms);
 - Cryptography requires computational problems that are hard.
 - Evolutionary computation courses

➤ From lecture 0: Tentative Schedule

No	Topic
1	DFA & Regular Languages
2	NFA
3	Regular Expressions
4	Non Regular Languages
5	PDA & Context Free Languages
6	Non Context Free Languages
7	Turing Machines

No	Topic
8	Decidable & Recognisable Languages
9	Undecidable Languages & reducibility
10	P Complexity Class
11	NP Complexity Class
12	NP Completeness
13	Cook-Levin Theorem
14	Revision Class

► Your feedback

- What was good?
- What can be improved?

► Your feedback (2): Teaching evaluation

- Deadline: 5 June (24:00) (anonymous to teachers)
- **important** reference basis for teachers to do a good job in the continuous improvement of teaching effectiveness
- Two ways:
 - Web end: <https://tis.sustech.edu.cn/> -Business Processing - Evaluation Task -2026 Spring Semester Student Evaluation Task
 - WeChat end: Enter the "Southern University of Science and Technology" WeChat enterprise account - Teaching Quality Management Platform: "My Tasks - Pending Evaluation" section.
- 3 minutes to complete (all items mandatory)
- The Teaching Affairs Department will give course selection points to students who complete it, according to each student's objectivity in the evaluation questionnaire for each course



► CS-338 exam

- When?

Wednesday, 17th June (10:30 – 12:30)

智华楼 310

► What is relevant?

- Content from **all lectures** and **all labs** can come up.
- A good exam will cover all bases.
- Make sure you answer all the questions!
- **Lab exercises** are a good preparation.
- Mock exam available.